

GLOBAL DATA CENTER & AI WATER/ELECTRICITY USAGE

Name : Manikant Vilas Vengal

Student : AF05214665

Batch Code : ANP-D4853

Project Name : Global Data Center & Ai Water/Electricity Usage

Project Guide : Rashri Maam

Objective

The objective of this project is to analyze the operational performance of global data centers by examining electricity consumption, water usage, Power Usage Effectiveness (PUE), cooling systems, facility capacity, and geographical distribution. The dashboard aims to provide meaningful insights into resource utilization, sustainability, and energy efficiency using Microsoft Excel.

The project helps users:

- Analyze Electricity and Water consumption across data centers.
- Compare Power Usage Effectiveness (PUE) among facilities.
- Study the impact of different cooling systems.
- Evaluate facility capacity across countries.

Exploratory Data Analysis (EDA)

- Understanding the distribution of data centers worldwide.
- Studying daily water consumption.
- Analyzing Power Usage Effectiveness (PUE).
- Identifying facilities with the highest capacity.
- Examining ownership by different companies.

- Comparing cooling system technologies.
- Identifying high-resource-consuming facilities.

Data Visualization

- Representing global data center performance using KPI Cards, Bar Charts, Pie Charts, Area Charts, Doughnut Charts, and Horizontal Bar Charts.
- Creating an interactive Excel dashboard with slicers for Year, Facility Type, Cooling System, and Surrounding conditions to enable dynamic analysis.
- Comparing electricity usage, water consumption, Power Usage Effectiveness (PUE), Water Usage Effectiveness (WUE), facility capacity, and cooling system distribution through visual reports.

Performance Analysis

- Evaluating electricity usage, water consumption, and estimated facility capacity across different data center facility types.
- Comparing Power Usage Effectiveness (PUE), Water Usage Effectiveness (WUE), and cooling system performance to assess operational efficiency.
- Analyzing the impact of facility type, cooling systems, surrounding conditions, and yearly trends through interactive dashboard visualizations.

Dashboard Development

- Developing an interactive Microsoft Excel dashboard using Pivot Tables, Pivot Charts, KPI Cards, and Slicers to visualize global data center performance.
- Performing data cleaning, transformation, and aggregation to ensure accurate analysis of electricity usage, water consumption, facility capacity, PUE, and WUE metrics.

- Designing user-friendly visual reports that enable dynamic filtering by Year, Facility Type, Cooling System, and Surrounding Conditions for efficient data exploration and decision-making.

Problem Statement

Global data centers supporting AI and cloud computing consume significant amounts of electricity and water, making it challenging to monitor resource utilization and operational efficiency. Analyzing this data manually is time-consuming due to the large volume of information related to facility capacity, cooling systems, and performance metrics.

This project aims to use Microsoft Excel to analyze global data center operational data and present meaningful insights through an interactive dashboard by addressing the following challenges:

1. Importing, cleaning, and organizing the dataset to ensure accurate and consistent analysis.
2. Analyzing electricity usage, water consumption, facility capacity, Power Usage Effectiveness (PUE), and Water Usage Effectiveness (WUE) across different facility types and cooling systems.
3. Visualizing key performance indicators and operational trends using KPI Cards, Pivot Charts, and interactive slicers to support better understanding and data-driven decision-making.

By implementing this project, the objective is to develop an efficient and interactive Excel dashboard that simplifies the analysis of global data center resource utilization and provides actionable insights into energy efficiency and sustainability.

Data Description

Source

The dataset was obtained from Kaggle and contains operational information about global data centers, including electricity consumption, water usage, cooling systems, ownership, geographical location, and sustainability metrics.

Data Model

A data model defines the structure and relationships within the Global Data Center & AI Water/Electricity Usage dataset, ensuring efficient storage, analysis, and visualization of operational metrics.

1. Data Model Overview

The data model follows a structured tabular format where each record represents a single data center facility and contains information related to its location, ownership, facility type, cooling system, electricity usage, water consumption, capacity, and efficiency metrics. This structure enables effective analysis and the creation of interactive dashboards in Microsoft Excel.

2. Entities and Attributes

Attribute Name	Description
Year	Year of data observation
Facility_ID	Unique identifier for each data center facility
Facility_Name	Name of the data center
Owner_Company	Company operating or owning the facility
Country	Country where the facility is located
City	City where the facility is located
Facility_Type	Type of data center (Colocation, Enterprise/Standard, Hyperscale/AI)
Cooling_System	Cooling technology used (Air Cooled, Evaporative, Liquid Cooled)
Surrounding	Environmental condition around the facility (High, Medium, Low)
Estimated_Capacity_MW	Estimated power capacity of the facility in megawatts

Attribute Name	Description
Daily_Electricity_Usage_MWh	Daily electricity consumption in megawatt-hours
Daily_Water_Usage_Gallons	Daily water consumption in gallons
PUE	Power Usage Effectiveness of the facility
WUE_L_per_kWh	Water Usage Effectiveness (liters per kWh)

3. Data Relationships

- Each facility belongs to one owner company.
- Each facility is located in one city and one country.
- Every facility is categorized into one facility type.
- Each facility uses one cooling system.
- Every facility has one estimated capacity, PUE, and WUE value.
- Each record contains daily electricity and water usage metrics for performance analysis.
- The Year attribute enables trend analysis and comparison across different time periods.

Approach

1. Data Import

Software Used

- Microsoft Excel
- Microsoft Power BI Desktop

Process

- Import the Global Data Center & AI Water/Electricity Usage dataset into Microsoft Excel.
- Verify column names, data types, and data consistency.

- Prepare the dataset for analysis by organizing it into a structured tabular format.

2. Data Cleaning

- Check for missing or incomplete values.
- Remove duplicate records.
- Correct and standardize data types (text, numbers, dates).
- Validate facility names, country names, and categorical fields such as Facility Type and Cooling System.
- Prepare the dataset for accurate analysis and visualization.

3. Data Analysis and Visualization

An interactive dashboard was developed using Microsoft Excel to analyze the operational performance of global data centers.

The dashboard includes:

- KPI Cards displaying Total Electricity Usage, Total Water Usage, Average PUE, Average WUE, Total Facility Capacity, and Total Facilities.
- Bar Charts comparing electricity usage, water consumption, and facility capacity across different facility types.
- Pie and Doughnut Charts illustrating the distribution of cooling systems and surrounding conditions.
- Area Charts showing yearly trends in electricity and water usage.
- Interactive Slicers for filtering data by Year, Facility Type, Cooling System, and Surrounding Conditions to enable dynamic analysis.

Project Result

1. Data Collection and Preparation

The Global Data Center & AI Water/Electricity Usage dataset was imported into Microsoft Excel. The dataset was cleaned by removing duplicate records, handling missing values, verifying data types, and standardizing categorical fields to ensure accurate analysis.

2. KPI Dashboard

Displays:

- Total Electricity Usage
- Total Water Usage
- Average Power Usage Effectiveness (PUE)
- Average Water Usage Effectiveness (WUE)
- Total Estimated Facility Capacity
- Total Data Center Facilities

Result:

Provides a quick overview of the operational performance and resource utilization of global data centers.

3. Electricity Usage Analysis

Displays:

- Electricity Usage by Facility Type

Result:

Helps identify which types of data centers consume the highest amount of electricity.

4. Water Usage Analysis

Displays:

- Water Usage by Facility Type

Result:

Shows the distribution of water consumption across different data center facility types.

5. Capacity Analysis

Displays:

- Estimated Facility Capacity (MW) by Facility Type

Result:

Highlights facility types with the highest operational capacity and infrastructure scale.

6. Cooling System Analysis

Displays:

- Distribution of Cooling System Types

Result:

Provides insights into the adoption of different cooling technologies used by global data centers.

7. PUE and WUE Analysis

Displays:

- Average Power Usage Effectiveness (PUE)
- Average Water Usage Effectiveness (WUE)

Result:

Evaluates the energy and water efficiency of data center operations, helping identify opportunities for improving sustainability.

8. Yearly Resource Consumption Trend

Displays:

- Year-wise Electricity Usage
- Year-wise Water Usage

Result:

Illustrates trends in resource consumption over time and supports performance monitoring across different years.

9. Interactive Dashboard Analysis

Displays:

- Dynamic filtering using Year, Facility Type, Cooling System, and Surrounding Conditions slicers.

Result:

Enables users to interactively explore the dataset, compare operational metrics, and gain meaningful insights for informed decision-making regarding energy efficiency and sustainability.

Conclusion

This project provides a comprehensive analysis of global data center operations by examining electricity usage, water consumption, facility capacity, Power Usage Effectiveness (PUE), and cooling systems through an interactive Microsoft Excel dashboard. The dashboard transforms raw operational data into meaningful visualizations, enabling users to evaluate energy efficiency, sustainability, and infrastructure performance.

The project demonstrates practical data analytics techniques such as data cleaning, Pivot Tables, KPI creation, dashboard design, and interactive filtering. Future enhancements may include predictive analytics, real-time monitoring, carbon emission analysis, and integration with Business Intelligence tools such as Power BI.